

# MONTH 5 College Student Management System

Full-Stack Java (Spring Boot + React.js + MySQL + JWT)

## User Roles

| Role  | Permissions                                                                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Admin | :Full access to all modules: User Management, Student Management, Document Generation, Marks, Fee Management, Dashboard, etc.                                                    |
|       | :Access limited to Student info viewing, Document Generation, Marks Management, Teacher Fee Viewing/Update (limited), and Dashboard Summary. No user or student deletion rights. |

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## Project Objective

Develop a secure, scalable full-stack web application to efficiently manage student data, documents, marks, fees, and provide a dynamic dashboard for admins and teachers.

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## 1 User Management (Admin only)

| Field     | Type      | Description                 |
|-----------|-----------|-----------------------------|
| id        | Long      | Primary key, auto-generated |
| fullName  | String    | Full name of user           |
| email     | String    | Unique email for login      |
| password  | String    | Encrypted password          |
| role      | Enum      | ADMIN / TEACHER             |
| createdAt | Timestamp | Account creation time       |

API Endpoints:

- POST /api/register — Register Admin/Teacher
- POST /api/login — Login (returns JWT)
- GET /api/users/me — Get current user profile
- GET /api/users — List all users
- PUT /api/users/{id} — Update user info

- DELETE /api/users/{id} — Delete user
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## 2 Student Management

| Field      | Type      | Description                 |
|------------|-----------|-----------------------------|
| id         | Long      | Primary key, auto-generated |
| name       | String    | Full student name           |
| rollNumber | String    | Unique roll number          |
| department | String    | Department/Faculty          |
| year       | Integer   | Academic year (1,2,3,4...)  |
| email      | String    | Student email               |
| phone      | String    | Contact phone number        |
| address    | String    | Home address                |
| createdAt  | Timestamp | Record creation timestamp   |

### API Endpoints:

- GET /api/students — List all students
  - GET /api/students/{id} — Get student details
  - POST /api/students — Add new student (Admin only)
  - PUT /api/students/{id} — Update student (Admin only)
  - DELETE /api/students/{id} — Delete student (Admin only)
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## 3 Document Generation (Bonafide, Transfer Certificate, Marksheet)

| Field        | Type | Description                                            |
|--------------|------|--------------------------------------------------------|
| id           | Long | Primary key, auto-generated                            |
| studentId    | Long | Foreign key to Student                                 |
| documentType | Enum | BONAFIDE / TRANSFER_CERTIFICATE / MARKSHEET            |
| issueDate    | Date | Document issue date                                    |
| content      | Text | Certificate content / marksheet details (JSON or text) |

| Field     | Type      | Description                    |
|-----------|-----------|--------------------------------|
| createdBy | Long      | User ID who generated document |
| createdAt | Timestamp | Record creation timestamp      |

API Endpoints:

- POST /api/documents/bonafide — Generate bonafide
- GET /api/documents/bonafide/{id} — Get bonafide
- POST /api/documents/transfer-certificate — Generate transfer certificate
- GET /api/documents/transfer-certificate/{id} — Get transfer certificate
- POST /api/documents/marksheet — Generate marksheet
- GET /api/documents/marksheet/{id} — Get marksheet

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#### Marks Management

| Field         | Type      | Description                 |
|---------------|-----------|-----------------------------|
| id            | Long      | Primary key, auto-generated |
| studentId     | Long      | Foreign key to Student      |
| subject       | String    | Subject name                |
| marksObtained | Integer   | Marks obtained              |
| maxMarks      | Integer   | Maximum marks for subject   |
| semester      | Integer   | Semester number             |
| createdAt     | Timestamp | Record creation timestamp   |

API Endpoints:

- GET /api/marks — List all marks
- POST /api/marks — Add marks
- PUT /api/marks/{id} — Update marks
- DELETE /api/marks/{id} — Delete marks

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#### Fee Management

| Field         | Type      | Description                                  |
|---------------|-----------|----------------------------------------------|
| id            | Long      | Primary key, auto-generated                  |
| studentId     | Long      | Foreign key to Student                       |
| totalAmount   | Decimal   | Total fee amount                             |
| paidAmount    | Decimal   | Amount paid so far                           |
| dueAmount     | Decimal   | Amount remaining                             |
| paymentStatus | Enum      | PAID / PARTIAL / DUE                         |
| paymentDates  | JSON      | List of payment dates and amounts (optional) |
| createdAt     | Timestamp | Record creation timestamp                    |

#### API Endpoints:

- GET /api/fees — List fee records
- GET /api/fees/{studentId} — Fee details for a student
- POST /api/fees — Add fee record (Admin only)
- PUT /api/fees/{id} — Update fee/payment status
- DELETE /api/fees/{id} — Delete fee record (Admin only)

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#### 6 Dashboard (Role-based summary)

| Field / Metric               | Description                                                     |
|------------------------------|-----------------------------------------------------------------|
| totalStudents                | Count of all students                                           |
| totalFeesCollected           | Sum of all paid amounts                                         |
| totalFeesPending             | Sum of all due amounts                                          |
| pendingDocuments             | Number of pending document requests                             |
| averageMarksByDept           | Average marks grouped by department                             |
| recentActivities             | Recent changes like student adds, fee updates                   |
| teacherAssignedStudentsCount | Number of students assigned to the logged-in teacher (optional) |
| pendingMarksEntries          | Count of marks yet to be entered by teacher                     |

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## Security & Authentication

- JWT for authentication
  - Spring Security for role-based authorization
  - Frontend React Router protection using stored JWT token and role checks
  - Backend API role validation for every sensitive operation
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## Database Relationships Overview

- users (id) referenced by documents.createdBy
  - students (id) referenced by documents.studentId, marks.studentId, fees.studentId
  - Payment history (optional) stored in a JSON column or separate table linked to fees
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## Frontend Modules (React.js)

- Login & Registration (with role selection)
  - Dashboard (admin & teacher views)
  - User Management (Admin only)
  - Student Management (Admin full CRUD, Teacher read-only)
  - Document Generation & Viewing (Both roles)
  - Marks Management (Both roles)
  - Fee Management (Admin full, Teacher limited)
  - Student Search & Filters
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## Project Timeline (4 Weeks)

| Week Backend |                                                                  | Frontend                           |
|--------------|------------------------------------------------------------------|------------------------------------|
| 1            | User Auth + JWT + DB Setup                                       | Login/Register Pages               |
| 2            | CRUD for Students & Search API                                   | Student Pages (List/Add/Edit/View) |
| 3            | Documents, Marks, Fee APIs                                       | Document, Marks & Fee Pages        |
| 4            | Swagger Docs + Pagination + Tests Dashboard, UX polish, Finalize |                                    |